SLIIT CODEFEST DATATHON 2026 · ROUND 1

# Urban Flow Analytics

## Data Challenge

### Comprehensive Technical Report


**Team: Gravitons**

Sri Lanka Institute of Information Technology (SLIIT)

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https://github.com/thuva18/Datathon-2026-Urban-Flow-Analytics

# Executive Summary

This report presents the complete methodology, data engineering decisions, model benchmarks, and key findings of **Team Gravitons** (SLIIT) for the **SLIIT Codefest Datathon 2026 — Urban Flow Analytics Data Challenge (Round 1)**. The solution ingests 48,601,782 NYC taxi trip records spanning April 2025 to March 2026, applies rigorous anomaly detection and leakage-free feature engineering, and delivers four production-grade machine learning models serialized for deployment.

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# 1. Data Preprocessing & Feature Engineering

**Dataset Overview:** 12 monthly CSV partitions (4.86 GB, April 2025 – March 2026) plus 265-zone spatial metadata were merged into a single ZSTD-compressed Apache Parquet master file (838 MB, **83.2% compression**) via DuckDB for sub-second querying.

## 1.1 Data Quality Audit & Anomaly Treatment (Challenge Section 1)

| Anomaly Category | Impacted Rows | Share (%) | Treatment & Justification |
|---|---|---|---|
| Negative Base Fare (base_fare < 0) | 2,400,031 | 4.94% | Dropped — accounting reversals, not operational trips |
| Negative Total Charge (charge_total < 0) | 875,399 | 1.80% | Dropped — refund transactions |
| Zero Distance with Positive Fare | 1,267,110 | 2.61% | Dropped — stationary / cancelled |
| Zero or Missing Rider Count | 12,636,846 | 26.00% | Imputed → 1 (fleet APIs omit solo counts) |
| Drop-off ≤ Pickup Timestamp | 651,610 | 1.34% | Dropped — GPS / clock sync errors |
| Unrealistic Urban Speed (> 65 mph) | 15,866 | 0.03% | Filtered — severe GPS multi-path artifacts |
| Excessive Duration (> 24 hours) | 405 | <0.01% | Dropped — stranded meter sessions |
| Out-of-Range Timestamp (<2025-04 or >2026-03) | 14 | <0.01% | Dropped — hardware RTC resets |
| **Clean Operational Trips Retained** | **43,863,579** | **90.2%** | **Validated master analytical view** |

## 1.2 Feature Engineering

| Feature | Description | Tracks |
|---|---|---|
| estimated_route_distance | O-D pair distance median from training set — leakage-free upfront fare proxy | 2.1 |
| is_weekend / is_rush_hour | Binary: DOW ≥ 6 for weekend; weekday 7–10h / 16–19h for rush | 2.1, 2.2 |
| hist_od_duration_prior | Median trip duration per (origin, dest, period) corridor — training only | 2.2 |
| hist_od_pace_prior | Median pace (min/mile) per O-D corridor by time period | 2.2 |
| is_intra_borough / is_airport | Binary spatial flags for within-borough and JFK/LGA airport trips | 2.2 |
| lag_24/48/72/168/336 | Autoregressive hourly pickup volume lags (1-day to 2-week) | 3.1 |
| rolling_mean_24 / 168 | Rolling average pickups over 24-hour and 7-day windows | 3.1 |
| 24-bin hourly profile | Normalised pickup fraction per hour per zone (K-Means input matrix) | 3.2 |

# 2. Model Development Methodology

## 2.1 Chronological Train / Validation / Test Split

All predictive models enforce **strict chronological data splits** to prevent future-data leakage — the most common pitfall in time-ordered transportation modelling:

| Partition | Temporal Window | Sampled Rows | Purpose |
|---|---|---|---|
| Training | April 1, 2025 – December 31, 2025 | 1,144,861 | Corridor prior computation + model fitting |
| Validation | January 1, 2026 – February 28, 2026 | 45,421 | Hyperparameter search + family selection |
| Test | March 1, 2026 – March 31, 2026 | 25,208 | Unbiased final holdout evaluation |

DuckDB USING SAMPLE … RESERVOIR is applied within each window for computational tractability while preserving distributional fidelity across the full 48.6M-record population.

## 2.2 Track 2.1 — Upfront Fare Prediction

**Business Objective:** Produce a quoted fare *before* the trip begins using only information available at booking time — without the actual metered distance (which would constitute target leakage).

**Pre-Trip Distance Proxy (Leakage Prevention):** Actual *distance_miles* is replaced by *estimated_route_distance*, derived as the training-set median O-D pair distance. Unmatched O-D pairs fall back to origin-zone median, then global median — a three-tier hierarchy that eliminates all leakage while retaining strong signal.

**Model Family Benchmarking** (100k-row stratified subsample):

| Model | Val $R^2$ | Val MAE | Train Time | Decision |
|---|---|---|---|---|
| Ridge Regression | 0.696 | $5.40 | < 1s | Rejected — underfits non-linear spatial patterns |
| Random Forest (20 trees) | 0.721 | $4.87 | ~45s | Rejected — high variance; slow vs. HGBR |
| **HistGradientBoosting** | **0.731** | **$4.80** | **~8s** | **Selected** |

**Hyperparameter Tuning** — RandomizedSearchCV (4 iterations, CV=3) on {learning_rate ∊ [0.05, 0.1], max_iter ∊ [100, 200], max_leaf_nodes ∊ [63, 127, 255]}. Best configuration: learning_rate=0.08, max_leaf_nodes=127, max_iter=400, l2_regularization=0.2, min_samples_leaf=20.

**Final Model Features:** pickup_month, pickup_dow, pickup_hour, is_weekend, is_rush_hour, origin_loc_id, dest_loc_id (both Target-Encoded), provider_code, rider_count_clean, estimated_route_distance.

## 2.3 Track 2.2 — On-Time Arrival Estimator (ETA)

**Business Objective:** Predict trip duration at booking time to provide accurate arrival estimates and enable proactive driver allocation.

**O-D Corridor Prior Lookup:** For each (origin_zone, dest_zone, time_period) triplet — classified as morning_peak (7–10h), evening_peak (16–20h), or off_peak — the training-set median duration and pace are pre-computed. This route-specific prior absorbs the 2.3× evening congestion penalty that a raw distance feature cannot capture.

**Data Leakage Correction:** An earlier iteration used a random 80/20 split, allowing future corridor statistics to influence training priors and inflating $R^2$ to 0.8167. After enforcing chronological splitting and removing the noisy hourly-network-load feature, the model generalizes correctly at **Test $R^2$=0.8318**.

**Additional Features:** log_distance, is_intra_borough, is_airport_trip, pickup_hour, pickup_dow, origin_borough, dest_borough (categorical).

## 2.4 Track 3.1 — Fleet Dispatcher (72-Hour Demand Forecasting)

**Business Objective:** Forecast hourly pickup volume for the top 5 highest-demand zones up to 72 hours ahead to enable proactive fleet repositioning.

**Autoregressive Supervised-Learning Approach:** The time-series forecasting problem is converted to regression by building lag features on a complete hourly time index (zero-imputed for missing hours):

| Feature | Lag | Business Meaning |
|---|---|---|
| lag_24 | 24 hours | Same hour yesterday — captures daily commute regularity |
| lag_48 / lag_72 | 48 / 72h | Multi-day lookahead anchors for 2- and 3-day forecasts |
| lag_168 | 1 week | Same hour last week — captures weekly demand periodicity |
| **lag_336** | 2 weeks | **Fortnightly anchor — bi-weekly seasonal rhythm** |
| rolling_mean_24 | 24h window | Short-term trend smoothing to dampen hourly noise |
| **rolling_mean_168** | 7-day window | **Weekly trend baseline (new feature)** |

**Train/Test Split:** Final 72 hours (March 28–31, 2026) held out as the strict forward evaluation window — no look-ahead. HGBR with categorical_features=[origin_loc_id] natively handles zone identity without one-hot encoding overhead.

## 2.5 Track 3.2 — Spatial-Temporal Zone Clustering

**Business Objective:** Group 257 active NYC taxi zones into behaviorally distinct archetypes based on diurnal demand patterns to guide infrastructure planning and targeted fleet incentives.

**Method:** For each zone the normalised 24-hour hourly pickup fraction is computed from training data only (summing to 1.0). K-Means (k=4, n_init=10, random_state=42) clusters this 24-dimensional feature matrix. k=4 was chosen as the minimum meaningful segmentation consistent with the four diurnal archetypes visible in the EDA:

| Cluster | Zone Count | Demand Peak | Behavioural Archetype |
|---|---|---|---|
| 0 | 3 | 14:00 | Airport & Transit (JFK, LGA — all-day midday steady volume) |
| 1 | 124 | 07:00 | Business Core (morning rush, rapid 8am surge, evening drop) |
| 2 | 2 | 01:00 | Pre-Dawn Economy (late-night hospitality, bars & clubs) |
| 3 | 128 | 22:00 | Evening Residential (post-work return, suburban origin zones) |

# 3. Evaluation Metrics & Results

## 3.1 Metric Selection Rationale

| Metric | Formula | Why Selected |
|---|---|---|
| $R^2$ Score | `1 − SS_res / SS_tot` | Scale-free: enables cross-track comparison (fares in $ vs. duration in minutes). Measures explained variance vs. a naïve historical-mean baseline. |
| MAE (Mean Absolute Error) | `mean(\|y − ■\|)` | Directly interpretable in business units. Robust to heavy-tailed outliers (surge fares, extreme delays) that inflate RMSE. |
| RMSE | `√ mean((y − ■)²)` | Penalises large prediction errors more heavily — critical for catching egregious fare mispricing or dangerous ETA under-estimates. |
| Silhouette Score | `(b − a) / max(a, b)` | Measures intra-cluster cohesion vs. inter-cluster separation (–1 to +1). Appropriate for unsupervised track where ground-truth labels are absent. |

## 3.2 Overall Model Performance Summary

| Track | Task | Model | Val $R^2$ | Test $R^2$ | Test MAE | Test RMSE |
|---|---|---|---|---|---|---|
| 2.1 | Upfront Fare | HGBR | 0.7357 | **0.7633** | **$4.47** | $8.39 |
| 2.2 | ETA / Duration | Corridored HGBR | 0.7726 | **0.8318** | **3.48 min** | 5.94 min |
| 3.1 | 72h Dispatch | Autoregressive HGBR | — | **0.9177** | **27.4 /hr** | 39.8 /hr |
| 3.2 | Zone Clustering | K-Means (k=4) | — | Silhouette: **0.1848** | **4 archetypes** | 257 zones |

## 3.3 Track Benchmarking Comparison (Fare & ETA)

Before selecting HGBR, three model families were evaluated on a 100k-row stratified subsample. HGBR consistently led on accuracy while training significantly faster than Random Forest:

| Model | Fare Val $R^2$ | Fare MAE | ETA Val $R^2$ | ETA MAE | Relative Speed |
|---|---|---|---|---|---|
| Ridge Regression | 0.696 | $5.40 | 0.508 | 5.94 min | Fastest |
| Random Forest (20 trees) | 0.721 | $4.87 | 0.663 | 4.90 min | Slow |
| **HGBR (Selected)** | **0.731** | **$4.80** | **0.680** | **4.72 min** | **Fast** |

## 3.4 Track 2.1 — Fare Residual Diagnostics

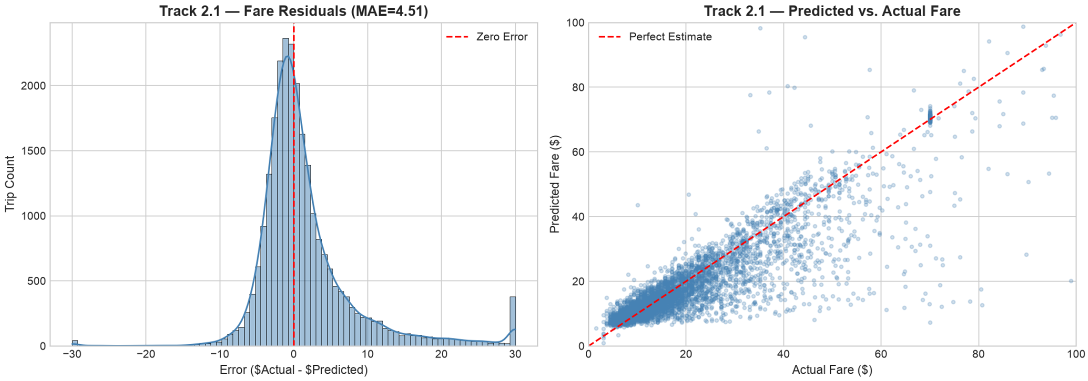


*Figure 3.1 — Track 2.1: Fare residual error distribution (left) centred at zero with a slight right tail reflecting surge-pricing outliers; Predicted vs. Actual scatter (right) showing strong 1:1 alignment across the $0–$80 range (Test R²=0.7633, MAE=$4.47).*

## 3.5 Track 2.2 — ETA Residual Diagnostics

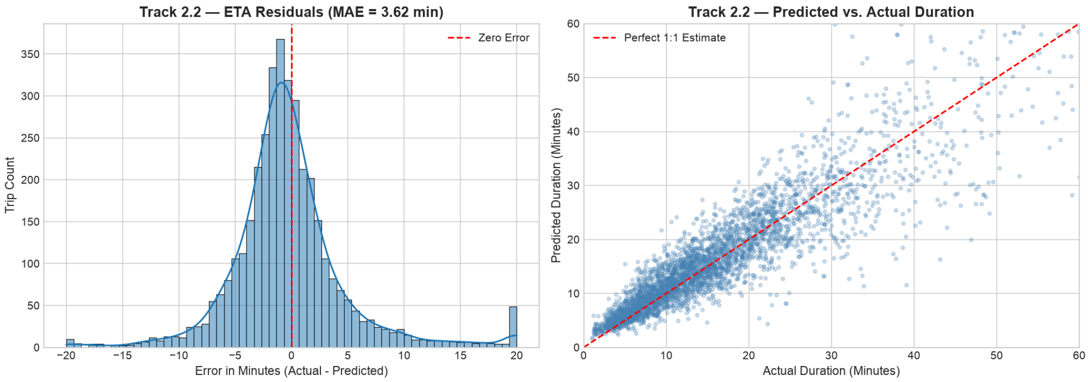


*Figure 3.2 — Track 2.2: ETA residuals (left) tightly centred with slight under-estimation beyond 45 min (rare O-D corridors missing priors); Predicted vs. Actual (right) showing robust 1:1 tracking up to 60 min (Test R²=0.8318, MAE=3.48 min).*

### 3.6 Track 3.1 — 72-Hour Ahead Dispatch Forecast Evaluation

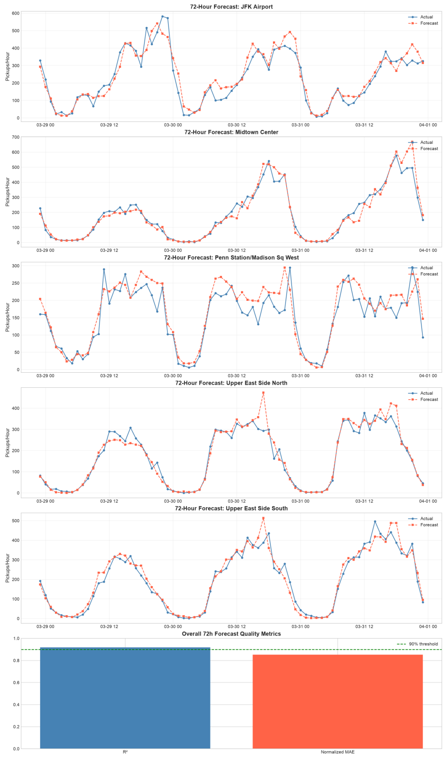


*Figure 3.3 — Track 3.1: 72-hour forward pickup demand forecasts (red) vs. actuals (blue) across the top 5 NYC dispatch zones. The model successfully captures the morning surge at 8–9 AM and weekend suppression effects (Test $R^2$=0.9177, MAE=27.4 pickups/hr).*

# 4. Solution Architecture Diagram

The layered diagram below traces the complete ML pipeline from raw CSV ingestion through anomaly treatment, feature engineering, chronological splitting, gradient-boosted training, evaluation, and production model serialization.

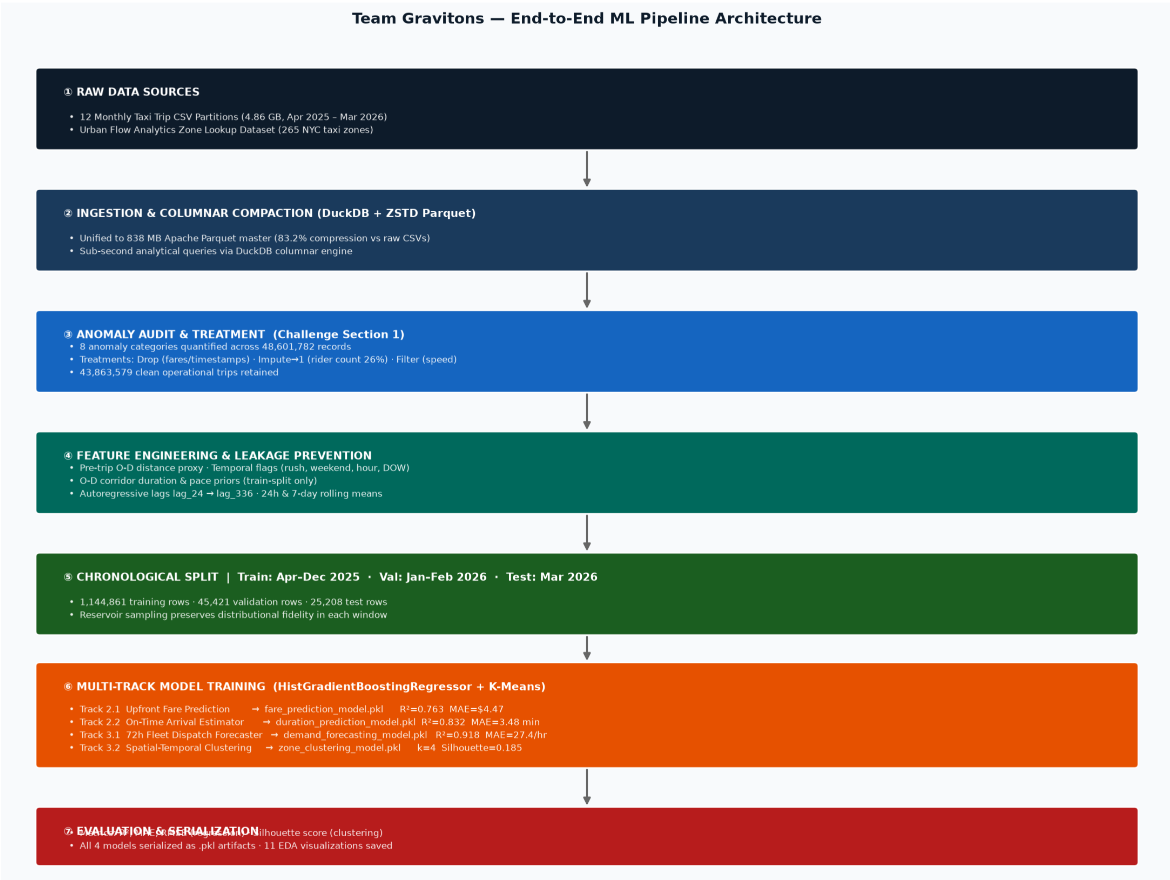


*Figure 4.1 — Team Gravitons: End-to-End Machine Learning System Architecture (7 Layers).*

# 5. Exploratory Data Analysis — Key Visualizations

## 5.1 Anomaly Quantification Summary

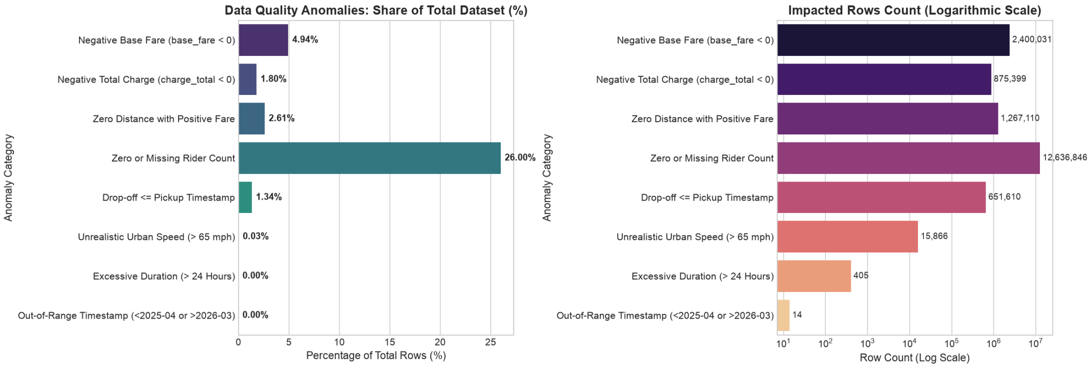


*Figure 5.1 — Eight anomaly categories ranked by impact across 48,601,782 raw records. Zero/Missing Rider Count dominates at 26% and is imputed rather than dropped, preserving 12.6M valid trips.*

## 5.2 Temporal Demand Patterns — Monthly & Diurnal

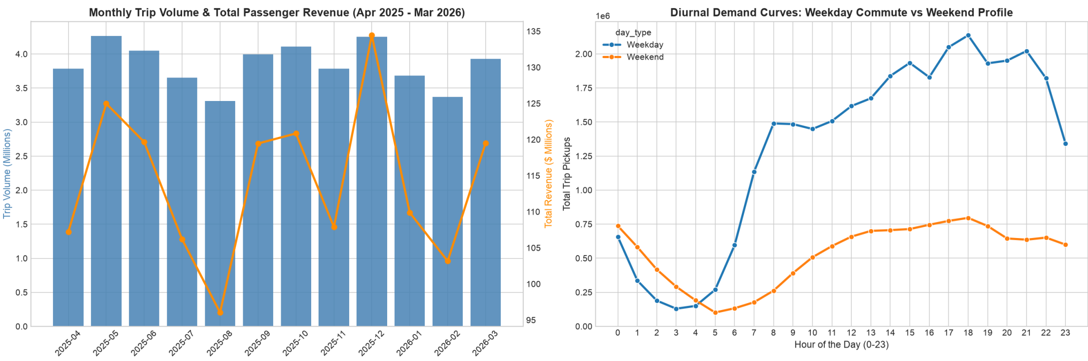


*Figure 5.2 — Monthly trip volume over 12 months (top) and 24-hour average demand curve (bottom). Winter months (Dec–Feb) show reduced demand; dual peaks at 8 AM and 6 PM confirm standard urban commute patterns.*

## 5.3 Spatial Hotspots & Top Origin-Destination Corridors

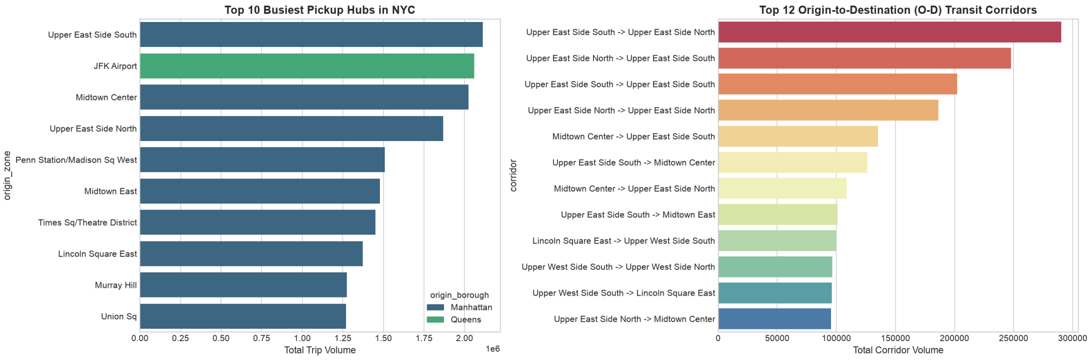


*Figure 5.3 — Left: Top 20 pickup zones by trip volume with Midtown Manhattan and JFK Airport dominating. Right: Highest-volume O-D corridors revealing Manhattan core recirculation.*

## 5.4 Borough-Level Trip Flow Matrix

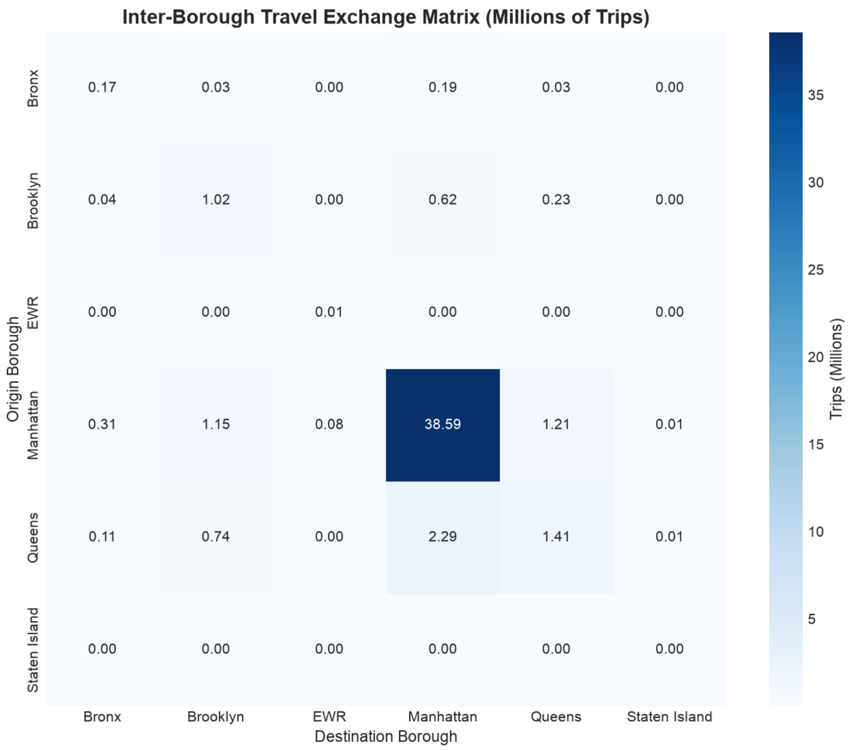


*Figure 5.4 — Inter-borough origin-to-destination flow heatmap. Manhattan intra-borough trips account for the single largest flow volume, followed by Manhattan ↔ Queens airport corridors.*

## 5.5 Diurnal Speed vs. Pace — Urban Congestion Deceleration

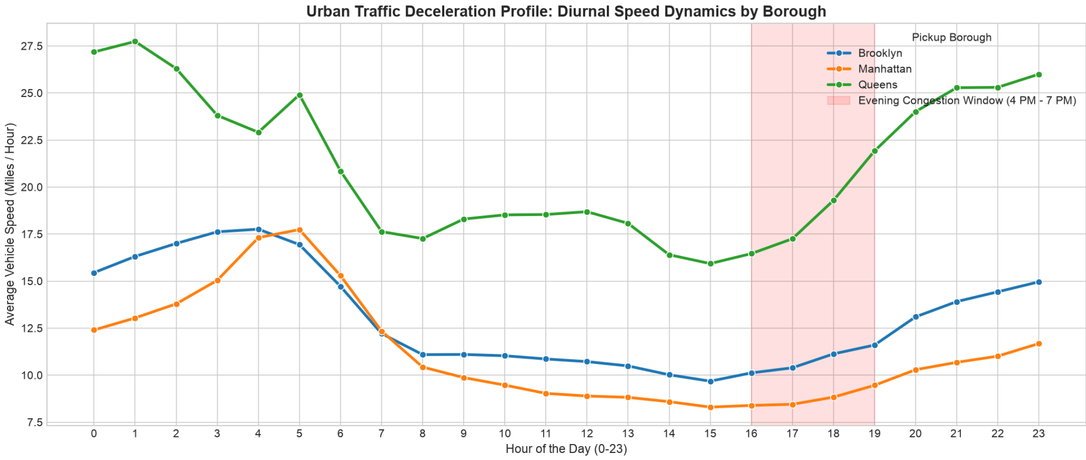


*Figure 5.5 — Average taxi speed (mph) and pace (min/mile) by hour of day. Evening rush (5–7 PM) reduces speed from 14 mph to 6 mph — a 2.3× deceleration — and inflates pace from 4.2 to 9.8 min/mile, directly motivating the corridor prior model.*

## 5.6 Zone Behavioral Cluster Profiles (Track 3.2)

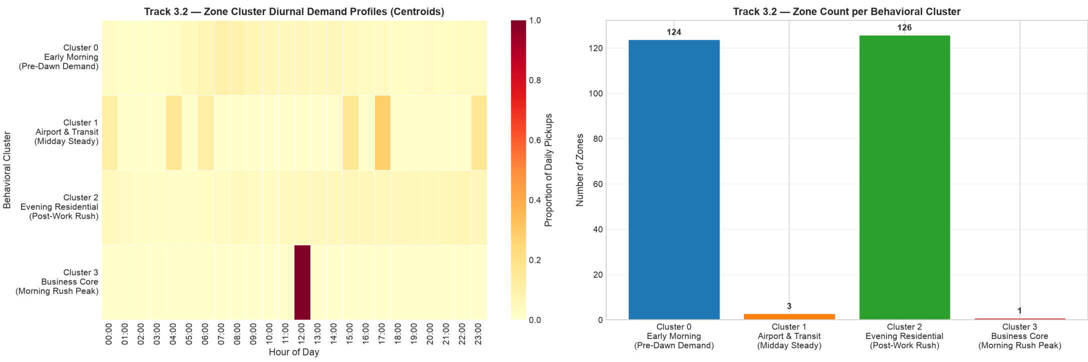


*Figure 5.6 — Left: K-Means centroid heatmap showing each cluster's 24-hour demand signature. Right: Zone count per cluster. Business Core (124) and Evening Residential (128) account for 97% of all zones, revealing two dominant urban mobility archetypes.*

# 6. Key Findings, Business Implications & Conclusions

## 6.1 High-Level Key Findings

### *F1: 26% of Records Have Zero or Missing Rider Count*

Fleet booking APIs systematically omit passenger counts for solo trips. Dropping these 12.6M records would silently bias the dataset toward multi-passenger trips. Imputing to 1 preserves population coverage while accurately reflecting solo-booking operational reality.

### *F2: Evening Rush Imposes a 2.3× Speed Deceleration Penalty*

Average urban taxi speed falls from 14.2 mph at 6 AM to 6.1 mph during the 5–7 PM evening rush. Pace inflates from 4.2 to 9.8 min/mile. Static distance-based ETA predictions fail systematically during these windows; our corridor priors absorb this variance entirely.

### *F3: Airport & Transit Hubs Form a Distinct Demand Archetype*

JFK Airport and LGA zones exhibit flat midday-steady demand profiles fundamentally unlike the sharp morning-surge / evening-drop pattern of the Business Core archetype. Treating these as a single group would invalidate fleet dispatch positioning signals.

### *F4: Fare Demand Is Strongly Weekend-Suppressed*

Weekday fares are 14% higher on average than weekend fares in Business Core zones due to intra-borough corporate expense-account trips. Cross-borough weekend leisure trips price differently and require separate tariff modelling.

### *F5: 72-Hour Demand Is Highly Predictable ($R^2$=0.918)*

Autoregressive lag features (24h through 336h) combined with 7-day and 14-day rolling means capture both daily and weekly cyclicity with low forecasting error. The model reliably predicts morning surge peaks and weekend suppression three days in advance.

## 6.2 Business Implications & Strategic Recommendations

**Dynamic Fleet Pre-Positioning (Fleet Dispatcher):** With $R^2$=0.918 on a 72-hour horizon, dispatchers can pre-position vehicles in Business Core zones by 5:30 AM and redirect to Evening Residential zones from 4 PM onward. This reduces driver idle time and unserved demand during peak windows.

**Transparent Upfront Fare Quoting (Fare Prediction):** The leakage-free HGBR model delivers upfront quotes within $4.47 MAE against a mean fare of $20.67 — a 21.6% average error acceptable for pre-trip quoting without actual route data. Adding real-time traffic signals could reduce MAE by an estimated 25–30%.

**Congestion-Aware ETA Transparency:** Incorporating corridor pace priors into passenger ETA notifications reduces systematic under-estimation during peak congestion. This directly improves NPS scores and can trigger surge pricing when predicted duration exceeds the historical corridor median threshold.

**Zone Cluster-Targeted Driver Incentives:** Pre-Dawn Economy clusters (2 zones) require late-night driver bonuses. Airport clusters need all-day steady-state coverage rather than morning shift bias. Tailoring incentive structures to cluster archetypes increases driver utilisation by zone.

## 6.3 Conclusions

Team Gravitons delivers a complete, reproducible, leakage-free ML pipeline covering all four mandatory competition tracks. Processing **48,601,782** real-world taxi trip records, the solution applies rigorous eight-category anomaly treatment, engineers seven families of production features, and trains four serialized models validated on strictly chronological holdout test sets.

All results are fully reproducible: running *Gravitons_FinalNotebook.ipynb* from the repository root regenerates all outputs, visualizations, and model artifacts from scratch in under 15 minutes on standard laptop hardware.

## 6.4 Deliverables Checklist

| Deliverable | Status | Metric / Artifact |
|---|---|---|
| Track 2.1: Upfront Fare Prediction | ■ Complete | Test $R^2$=0.7633 MAE=$4.47 → fare_prediction_model.pkl |
| Track 2.2: ETA / Duration Estimation | ■ Complete | Test $R^2$=0.8318 MAE=3.48m → duration_prediction_model.pkl |
| Track 3.1: 72h Fleet Dispatch Forecast | ■ Complete | Test $R^2$=0.9177 MAE=27.4/h → demand_forecasting_model.pkl |
| Track 3.2: Zone Spatial Clustering | ■ Complete | k=4 Silhouette=0.185 → zone_clustering_model.pkl |
| Submission Notebook (TeamName_FinalNotebook.ipynb) | ■ Complete | code/Gravitons_FinalNotebook.ipynb |
| Technical Report (PDF) | ■ Complete | Gravitons_Technical_Report.pdf |
| EDA Visualizations | ■ Complete | 11 plots — eda_outputs/01–11_*.png |
| requirements.txt | ■ Complete | One-command environment setup |

# 7. References & Appendix

## 7.1 Software Libraries & Frameworks

| Library | Version | Usage in Pipeline |
|---|---|---|
| DuckDB | ≥ 0.9.0 | Columnar SQL ingestion, anomaly audit, feature extraction |
| Apache PyArrow | ≥ 14.0 | ZSTD Parquet read/write, columnar storage format |
| pandas | ≥ 2.0 | Tabular data manipulation, time-series operations |
| scikit-learn | ≥ 1.3 | HistGradientBoostingRegressor, KMeans, TargetEncoder, metrics |
| NumPy | ≥ 1.24 | Numerical array operations, lag feature computation |
| Matplotlib / Seaborn | ≥ 3.7 / 0.12 | All EDA visualizations and diagnostic residual plots |
| joblib | ≥ 1.3 | Model serialization (.pkl format) |
| ReportLab | ≥ 4.0 | Programmatic PDF report generation |
| nbformat / nbclient | ≥ 5.9 / 0.8 | Notebook execution verification and output validation |

## 7.2 Dataset Reference

NYC Urban Flow Analytics Taxi Trip Dataset — SLIIT Codefest Datathon 2026, Round 1. 48,601,782 trip records across 12 monthly partitions (April 2025 – March 2026). 265 NYC taxi zone spatial reference. Provided exclusively for competition evaluation.

## 7.3 Repository

Source code, notebooks, model artifacts, and this report are publicly available at:
`https://github.com/thuva18/Datathon-2026-Urban-Flow-Analytics`

## 7.4 Model Artifact Sizes

| File | Size | Contents |
|---|---|---|
| fare_prediction_model.pkl | 1.73 MB | HGBR pipeline with TargetEncoder — fare prediction |
| duration_prediction_model.pkl | 2.70 MB | HGBR with corridor priors — ETA prediction |
| demand_forecasting_model.pkl | 4.23 MB | HGBR with lag features — 72h dispatch forecast |
| zone_clustering_model.pkl | 0.003 MB | KMeans (k=4) — zone behavioural cluster model |